GLUCOAHEAD

Two streams. One evolving patient twin.

Offline training and browser inference are separate. All supplied records are synthetic.

01 / Synthetic inputs

Static EHR profile

Age, BMI, HbA1c, duration

Dynamic CGM + wearable history

5-minute glucose, heart rate, steps

02 / Offline training

Patient-disjoint 80 / 20 / 30 split

Training-only standardization

Ridge regression predicts 12 deltas

Validation residuals set band

03 / Exported artifacts

model.json: learned coefficients

patients.json: held-out demo cases

metrics.json: exact benchmark

Generator and MIT-licensed source

JSON artifacts enter the static browser app

04 / Replay a sensor stream

Browser loads a synthetic profile

New reading advances virtual time

Last 60 minutes form input window

CSV import stays in tab memory

05 / Local model inference

core.js matches NumPy inference

EHR and sensor features fuse

12 forecasts: +5 to +60 minutes

Future readings never enter model

06 / Doctor-facing dashboard

History, forecast and residual band

Reference-crossing forecast note

Frozen prediction vs actual replay

Exportable JSON patient snapshot

Verification: NumPy/JS parity, causal input boundary, EHR and sensor sensitivity, CSV validation.

Engineering demonstration only. No real clinical records, device connection or treatment recommendation.